EE650 · BASICS OF MODERN CONTROL SYSTEMS · PROGRESS REVIEW

Fault-tolerant control of a quadrotor after complete rotor loss

Residual Koopman MPC, and certified limits on fault-detection delay

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AIM

One aim, two connected parts

MODEL AND CONTROL · AS PROPOSED

Residual Koopman MPC

Can a Koopman (EDMDc) model of only the mismatch, built on the known spinning-equilibrium model of Mueller & D'Andrea, predict and control better than the physics model alone or a full-state Koopman model?

TIMING · ADDED AFTER THE PROPOSAL

Certified detection-delay budget

A fault detector needs 30–130 ms to notice a dead rotor. What is the largest delay for which recovery is guaranteed, and how does the capture controller change it?

The link: each capture controller (LQR, analytical MPC, residual Koopman MPC) is compared by the detection delay it can tolerate.

THE PROBLEM

What happens when a rotor fails

Healthy flight
Normal controller, four rotors

→

Rotor fails
t = 0, thrust of rotor 4 lost

→

Undetected, τ
Healthy controller still flying a 3-rotor vehicle

→

Capture
Failure controller takes over

→

Relaxed hover
Steady spin ≈ 19 rad/s, axis tilted

Most recovery methods with a guarantee assume τ = 0.

Stability with a fixed delay is shown for a linear model; a certified maximum delay under thrust limits was not found.

Simha et al. 2017; Al Lawati & Akhtar 2026; Başak & Prempain 2015; Strack van Schijndel et al. 2021

FOUNDATION

Simulator and vehicle

- 6-DOF rigid body, quaternion attitude, rotor gyroscopic terms
- Per-rotor thrust 0.2–3.8 N; a motor can also be stopped
- First-order motor lag, 15 ms
- Reduced attitude $n = R^T e_3$: the "up" direction seen from the body
- RK4 integration; controller at 1–5 ms (zero-order hold)
- Truth plants for learning add thrust loss at high spin, inertia error and a different motor lag

Parameter	Value
Mass	0.50 kg
Arm length	0.17 m
Inertia Ixx, Izz	3.2, 5.5×10^{-3} kg m²
Thrust coefficient κf	6.41×10^{-6} N s²/rad²
Torque ratio κτ	0.0169 m
Yaw drag γ	2.75×10^{-3} N m s/rad

All values from Mueller & D'Andrea, ICRA 2014, Sec. V-A

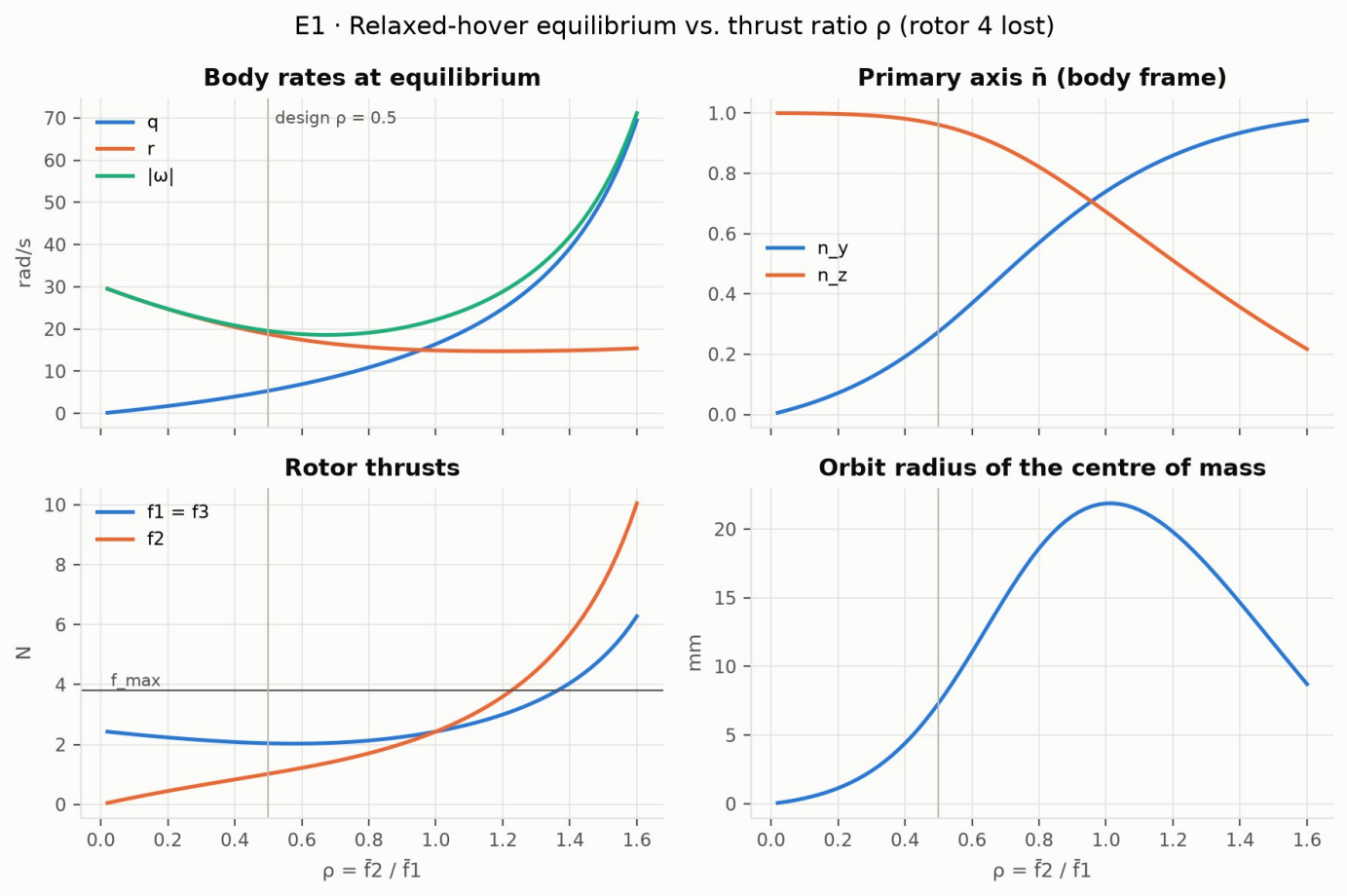
FOUNDATION

The model matches three published papers

Source	What we compared	Result
ICRA 2014	Two-rotor-loss equilibrium	All published digits (≤ 0.14%)
ICRA 2014	Three-rotor-loss equilibrium	All published digits (≤ 0.8%)
ICRA 2014	One-rotor-loss equilibrium	f, r, n_z ≤ 0.5%; pitch rate −6% *
ICRA 2014	ā and controllability loss (eqs. 28, 39)	Exact
IJRR 2016	Five failure cases, own implementation	Body rates within 4%
Sun et al. 2020	Bebop 2, two-rotor loss (eq. 60)	Exact
Flight data	Spin-up to 10 rad/s at the flight thrust (2.95 N)	**0.65 s vs 0.62 s in flight (4%)**

* The paper's own published values do not satisfy its eq. (12); its other two cases match exactly. **46 automated tests** also check conservation laws, closed-form solutions and these published numbers.

The spinning equilibrium



Rotor 4 lost, design ratio $\rho = f_2/f_1 = 0.5$:

f = (2.04, 1.02, 2.04) N
spin 19.6 rad/s, axis tilted 16°

Thrust, spin rate and tilt match the paper within 0.5%.

The ρ sweep reproduces the paper's Fig. 3 (minimum spin 18.6 vs 19.0 rad/s).

Linear model: controllable, but what can we observe?

7 states (p, q, r, n_x, n_y, z, v_z), inputs f_1, f_2, f_3. Linear and time-invariant in body axes.

Open-loop modes: ±10.1j, ±19.6j, −0.51 and an altitude double integrator.

Controllable: rank 7 of 7; every mode passes the PBH test.

Sensors	Rank	Missing
Gyro	3 / 7	tilt, altitude
Gyro + tilt estimate	5 / 7	altitude
Gyro + barometer	7 / 7	none (weak)
Gyro + tilt + barometer	7 / 7	none

The equilibrium is tilted, so vertical acceleration depends on tilt. A barometer alone then reveals the tilt, but about 2,700× more weakly than a tilt sensor (observability Gramian).

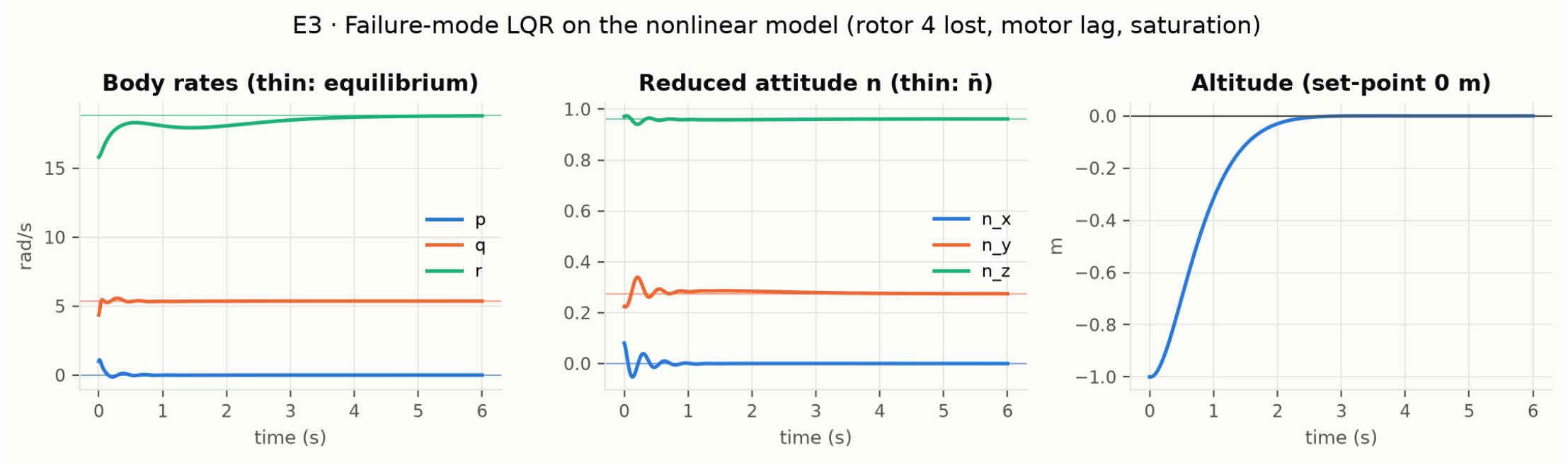
FOUNDATION

LQR baseline at the spinning equilibrium

18 ms
input-delay margin (healthy LQR: 40 ms)

ζ ≈ 0.19
damping of the least-damped mode

1.8–2.5 s
to settle altitude, tilt and yaw rate



Weights from Mueller & D'Andrea, plus motor states and altitude

PART 1 · AS PROPOSED

Residual Koopman models of the failed vehicle

Learn only what the physics model gets wrong, then use it inside a convex MPC.

SET-UP

Three predictors, same data

Analytical

Mueller & D'Andrea linear model about the spinning equilibrium. No data.

Full-state EDMDc

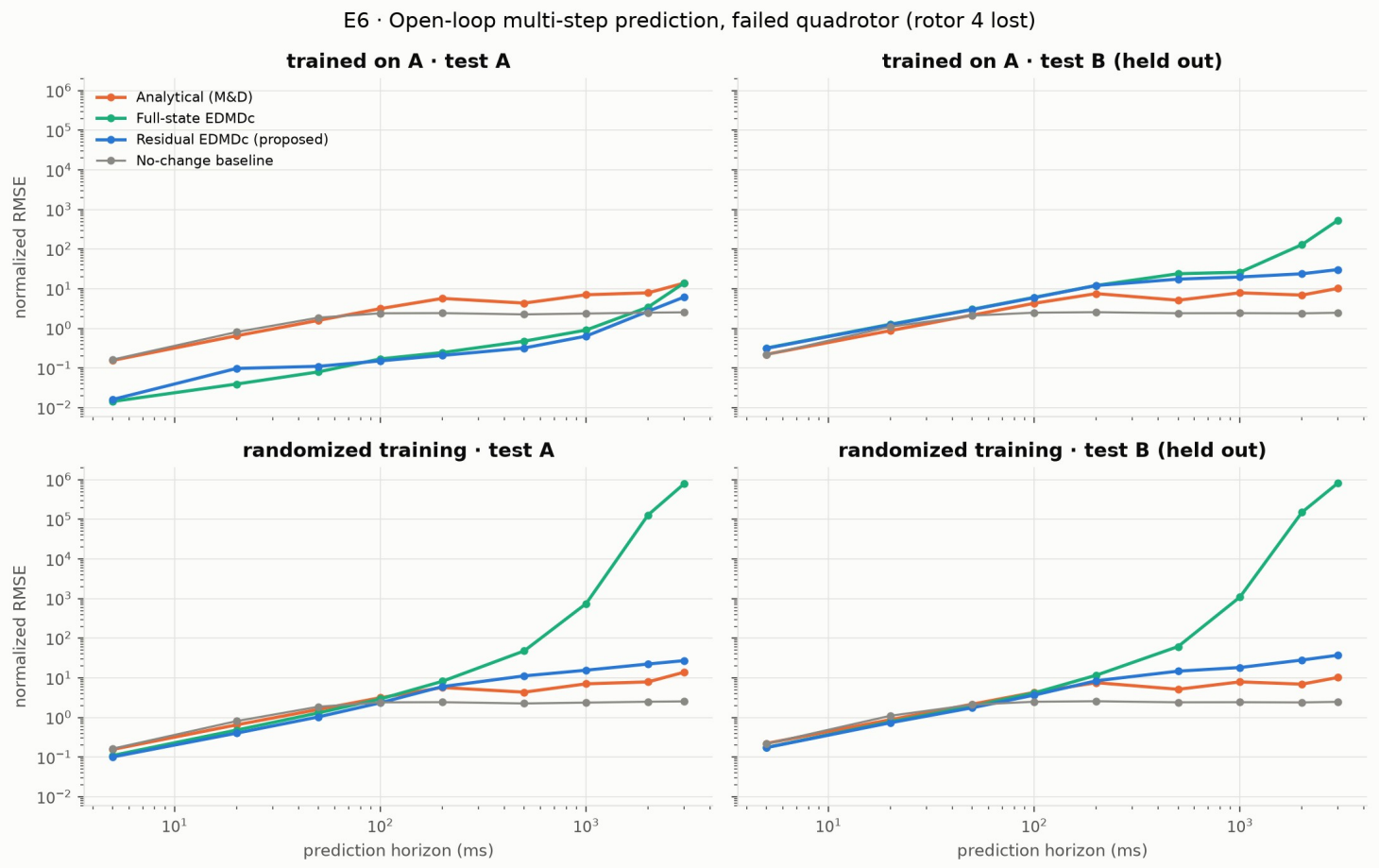
Learns all the dynamics from data: 55 lifted states (quadratic dictionary).

Residual EDMDc (proposed)

Physics model plus a learned one-step correction e = x_true − x_model, clipped at d_max.

Data: 40 episodes × 4 s at 5 ms, from a truth plant with effects the model lacks (thrust loss at high spin, inertia error, slower motors). **Test:** 10 new episodes from the same plant (A) and 10 from a different one (B).

From 100 ms on, the residual model predicts best



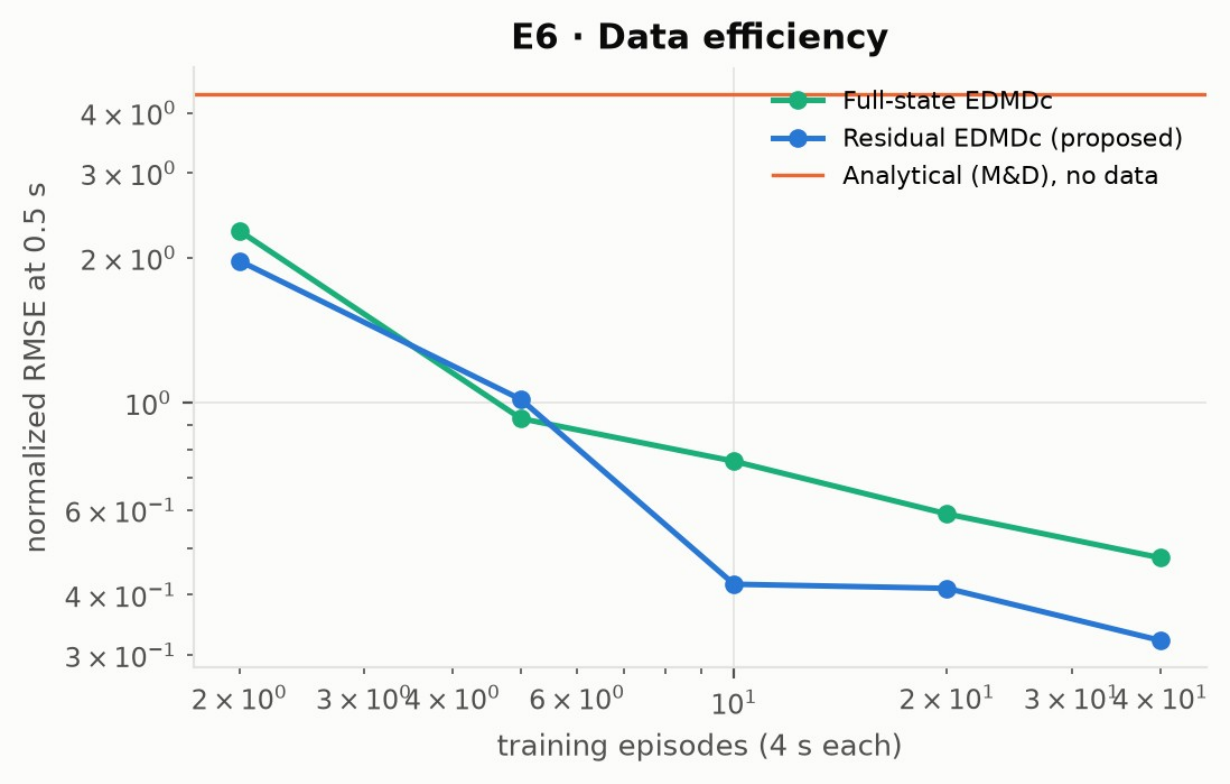
Error at a 1 s horizon, test family A (normalized RMSE):

Predictor	Error
Residual EDMDc	**0.64**
Full-state EDMDc	0.91
"No change" baseline	2.38
Analytical model	7.09

Below 100 ms the full-state model is slightly better. It has 4 of 55 eigenvalues outside the unit circle and falls behind "no change" by 2 s, as the proposal predicted.

LIMITS

Little data needed, no generalization yet



Data efficiency

With 10 episodes the residual model (0.42) beats the full-state model trained on 40 (0.48), at a 0.5 s horizon.

Generalization fails

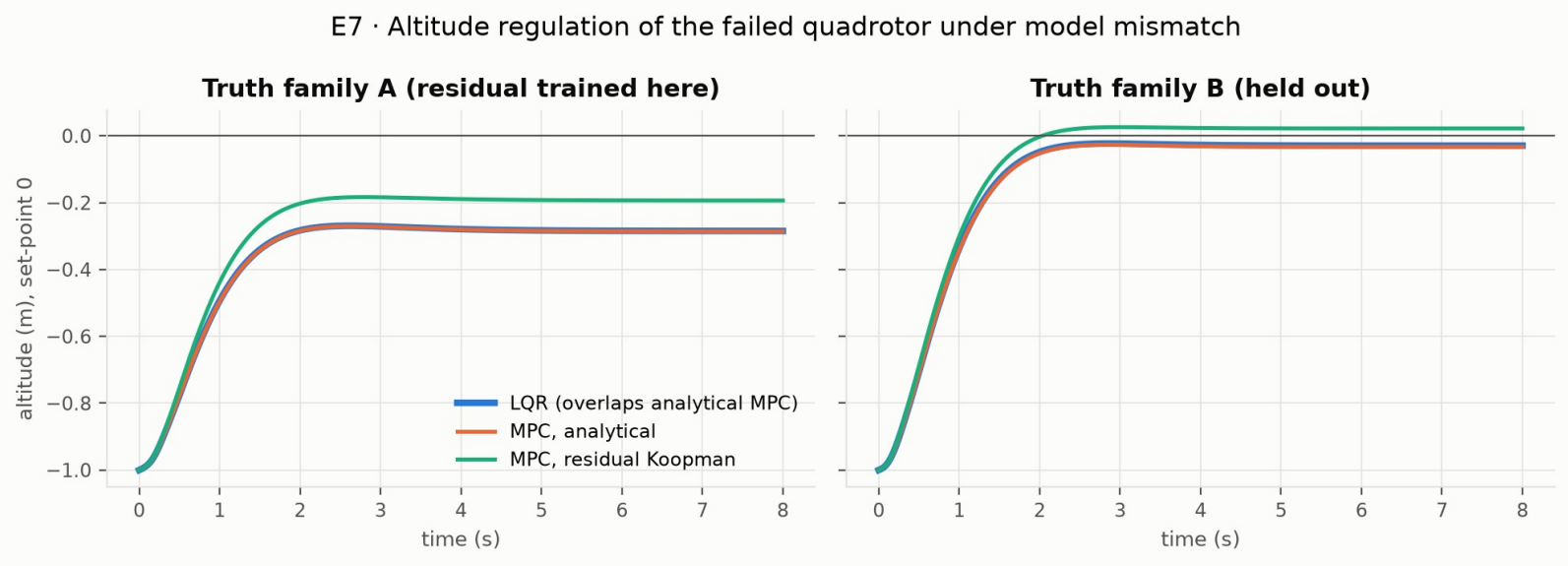
On the unseen plant B, both learned models are worse than the physics model alone. Training on randomized plants helps on B only up to about 100 ms and hurts on A: the unknown parameters cannot be read from one state sample.

Next

Update the residual online with recursive least squares (it is linear in its parameters).

CONTROL

Residual MPC removes a third of the bias



−32%

steady altitude error on plant A (−0.29 → −0.19 m)

0.7 ms

solver time per step (horizon 20 × 5 ms, OSQP)

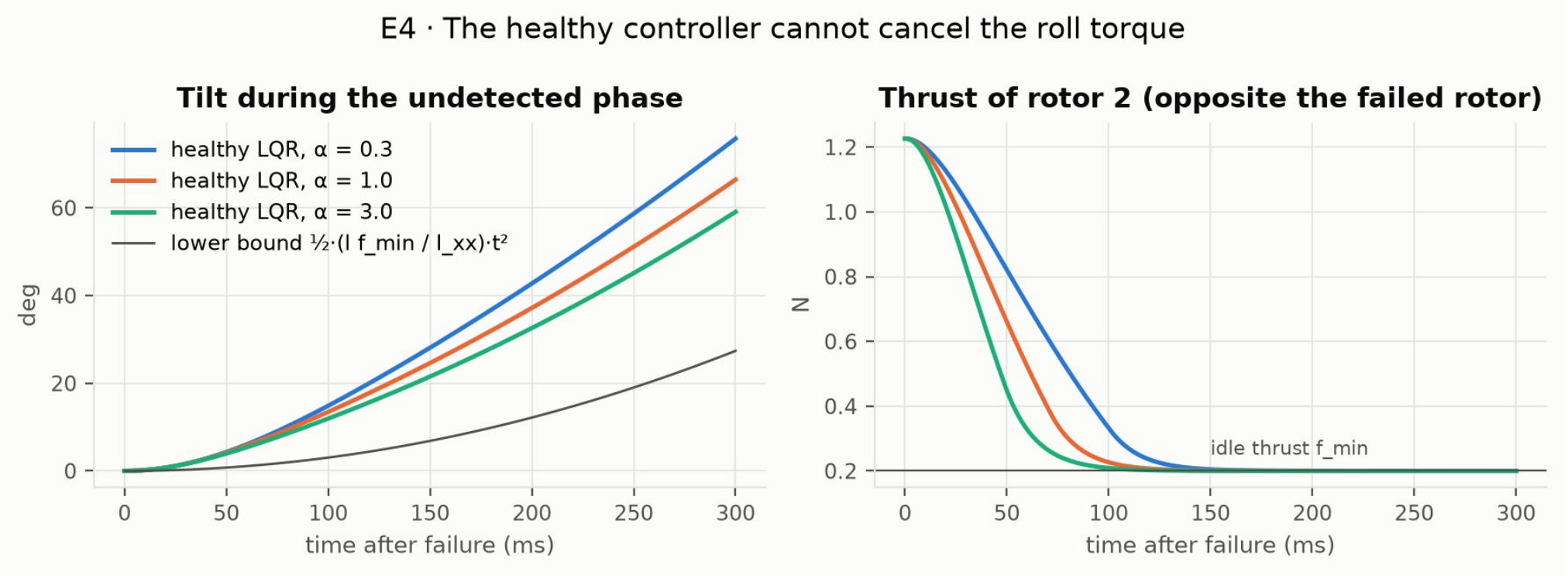
The residual enters the QP as a measured disturbance, so the problem stays convex (as in Todde et al. 2026). It helps for every mismatch effect of plant A, but on plant B without its thrust loss the error grows by 47%. After a detection delay, LQR is safe up to 10 ms and both MPCs only without delay.

PART 2 · ADDED AFTER THE PROPOSAL

How long can fault detection take?

Most recovery methods assume instant detection. Fast detectors take 30–130 ms.

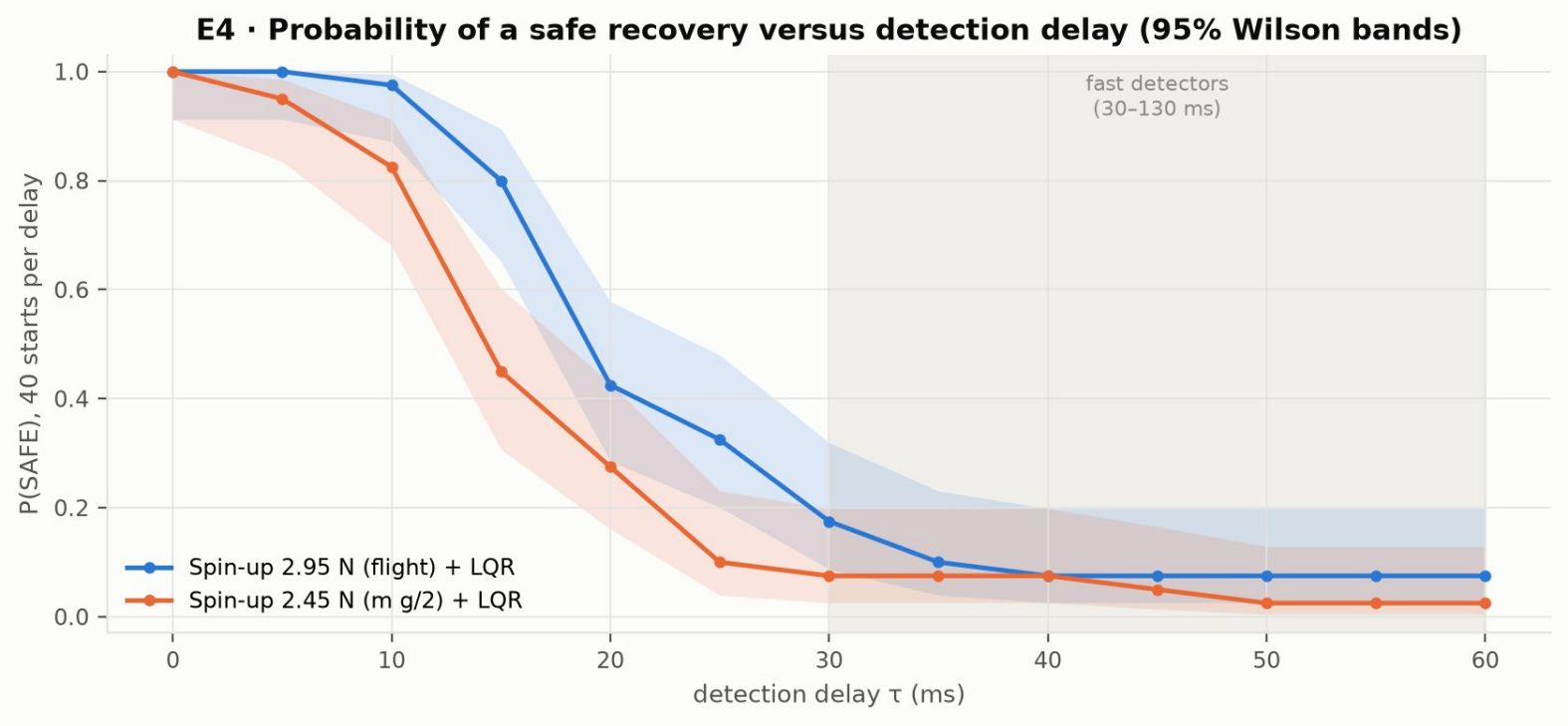
Before detection, nothing stops the roll



Rotor 2 cannot go below its idle thrust (0.2 N), so the roll acceleration is at least **l·f_min / I_xx = 10.6 rad/s²** for any healthy-controller tuning. Tilt reaches about 13° at 100 ms and 37° at 200 ms.

KEY RESULT

Safe recovery is likely only below about 12 ms



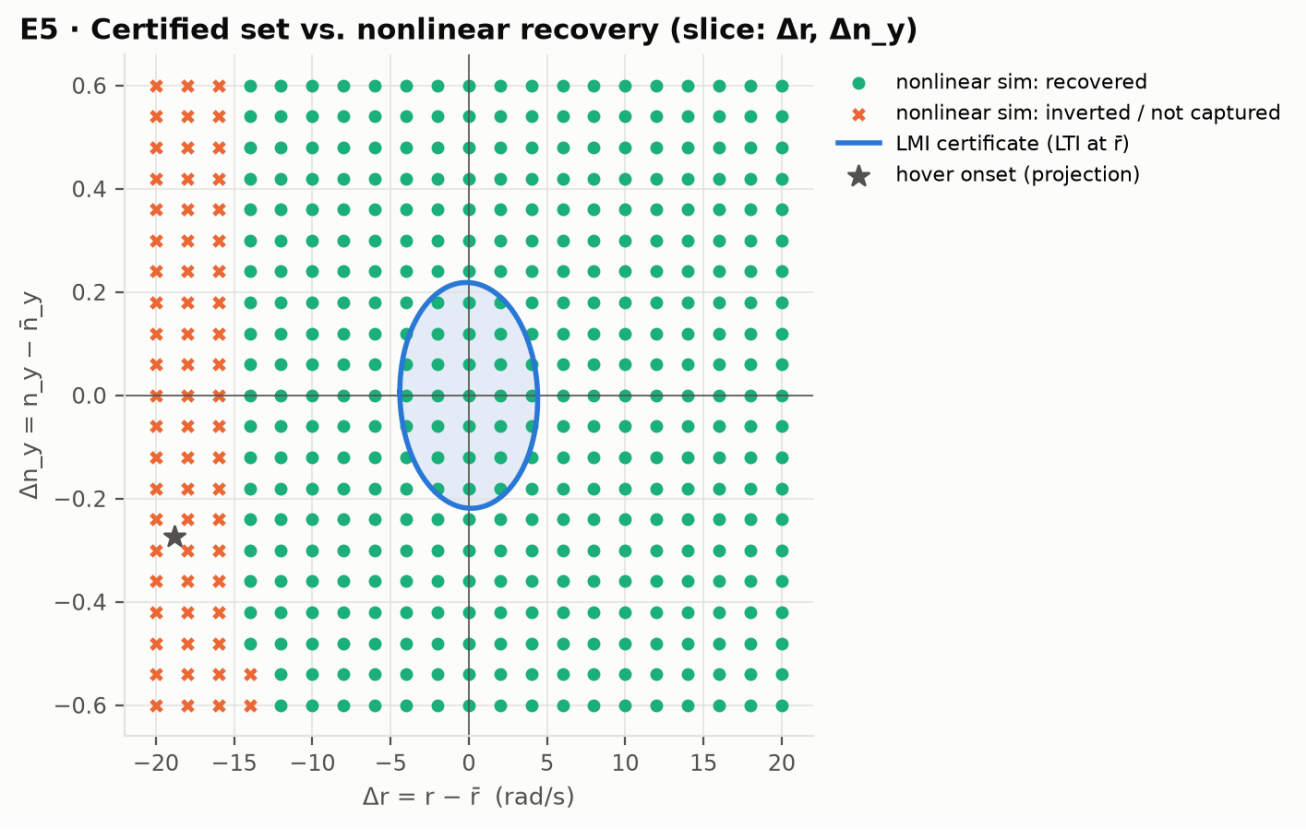
P(safe), 40 random starts per delay:

Delay	2.95 N	2.45 N
0 ms	1.00	1.00
10 ms	0.97	0.82
15 ms	0.80	0.45
20 ms	0.42	0.28
30 ms	**0.17**	**0.07**

2.95 N: flight spin-up thrust (Fig. 4 of the paper); 2.45 N = mg/2.

Safe = attitude recovered, altitude loss under 2 m, never upside down (fixed before the runs). With no delay the vehicle climbs 0.40 m and loses only 0.03 m, like the flight. P(safe) falls below 0.5 at 19 ms (14 ms at mg/2); for attitude capture alone, at 52 ms. Fast detectors need 30–130 ms.

A first LMI certificate, and its limit



Saturation-aware invariant ellipsoid (Hu–Lin LMIs) for the LQR on the linear model at the spinning equilibrium.

22%
of the way to hover onset, certified

70%
recovers in simulation

The LQR fails only at spin rates of 4.8 rad/s or less, which is where the hover-onset state lies. Next: a certificate valid over the whole spin-up, treating the yaw rate r as a varying parameter (LPV).

NEXT STEPS

Plan for the remaining weeks

Dates	Learning the mismatch	Detection-delay budget
12–18 Oct	MPC on the full-state Koopman model; integral-action MPC baseline	Read the remaining key papers; certificate over the spin-up (LPV in r)
19–25 Oct	Online residual adaptation (recursive least squares)	Reachable set of the undetected phase; first certified delay budget τ^*
26–31 Oct	Air-relative thrust in the truth plant	Sensitivity of τ^* to spin-up thrust, idle thrust and motor lag; presentation 31 Oct
1–13 Nov	Certificate for the residual-Koopman loop if time allows	Final simulations; journal-style report due 13 Nov

Main risk: a certificate over the whole spin-up may be infeasible or too conservative. Fallbacks: a parameter-dependent Lyapunov function, or a chain of nested ellipsoids.

SUMMARY

Where we stand

1 The simulator matches three published papers and the flight spin-up time within 4%.

2 The residual Koopman model predicts best from 100 ms on, needs less data and cuts the MPC altitude bias by a third, but does not yet generalize to an unseen vehicle.

3 The published recovery is safe with probability above 0.9 only up to about 12 ms of detection delay; detectors need 30–130 ms. Certifying the spin-up is next.

Questions?